Midterm Exam 2

55 minutes • 5 required problems + 1 bonus problem • 50 points + 6 bonus

Name: _____ **Student ID:** _____

Instructions. This exam is closed-book and closed-notes. Calculators are not allowed. Answer Problems 1–5. Problem 6 is optional bonus credit. For each problem, show your reasoning, including the steps in calculations. You may quote results proved in class unless a problem explicitly asks for a proof.

- **Your work must be neat. If I can't read or find it, I can't grade it.**
- **Partial credit will only be given for correct and substantial work.**
- **Please put away your electronic devices.**

Honor statement: I agree to take this exam in accordance with the University Honor Code. My signature on this exam is a contractual statement asserting all work to be my own, that I have not collaborated in any way with my classmates, that I have not used outside resources, including the Internet, and that I have no knowledge that any of my classmates have broken University regulations on this exam.

Sign and date below.

_____ _____

Problem	**1**	**2**	**3**	**4**	**5**	**6 (bonus)**	**Total**
Points	10	8	10	12	10	+6	**50 (+6)**
Score							

Problem 1 **10 points**

(a) *[6 points]* Use the Euclidean algorithm in $\mathbb{Z}[\sqrt{-2}]$ to compute

$$\gcd(5-4\sqrt{-2},\ -1+5\sqrt{-2})$$

up to multiplication by a unit.

Solution. Since $N(-1+5\sqrt{-2}) = 51 < N(5-4\sqrt{-2}) = 57$, we divide $5-4\sqrt{-2}$ by $-1+5\sqrt{-2}$ using the division algorithm in $\mathbb{Z}[\sqrt{-2}]$:

$$\frac{5-4\sqrt{-2}}{-1+5\sqrt{-2}} = \frac{(5-4\sqrt{-2})(-1-5\sqrt{-2})}{(-1+5\sqrt{-2})(-1-5\sqrt{-2})} = \frac{-45-21\sqrt{-2}}{51} = -\frac{15}{17} - \frac{7}{17}\sqrt{-2}.$$

Choose integers $x=-1$ closest to $-15/17$ and $y=0$ closest to $-7/17$. Then the remainder is

$$5-4\sqrt{-2}-(-1)(-1+5\sqrt{-2}) = 4+\sqrt{-2}.$$

Similarly, we divide $-1+5\sqrt{-2}$ by $4+\sqrt{-2}$:

$$\frac{-1+5\sqrt{-2}}{4+\sqrt{-2}} = \frac{(-1+5\sqrt{-2})(4-\sqrt{-2})}{(4+\sqrt{-2})(4-\sqrt{-2})} = \frac{6+21\sqrt{-2}}{18} = \frac{1}{3} + \frac{7}{6}\sqrt{-2}.$$

Choose integers $x=0$ closest to $1/3$ and $y=1$ closest to $7/6$. Then the remainder is

$$-1+5\sqrt{-2} - \sqrt{-2}\,(4+\sqrt{-2}) = 1+\sqrt{-2}.$$

Repeating, we divide $4+\sqrt{-2}$ by $1+\sqrt{-2}$:

$$\frac{4+\sqrt{-2}}{1+\sqrt{-2}} = \frac{(4+\sqrt{-2})(1-\sqrt{-2})}{(1+\sqrt{-2})(1-\sqrt{-2})} = \frac{6-3\sqrt{2}}{3} = 2-\sqrt{2} \in \mathbb{Z}[i],$$

so $1+\sqrt{-2} \mid 4+\sqrt{-2}$. Hence, $\gcd(5-4\sqrt{-2},\ -1+5\sqrt{-2}) = 1+\sqrt{-2}$.

(b) *[4 points]* Express the gcd that you found in part (a) as a linear combination of $5-4\sqrt{-2}$ and $-1+5\sqrt{-2}$ in $\mathbb{Z}[\sqrt{-2}]$.

Solution. The calculations in part (a) gives

$$\begin{aligned} 5-4\sqrt{-2} &= (-1)(-1+5\sqrt{-2}) + 4 + \sqrt{-2} \\ -1+5\sqrt{-2} &= \sqrt{-2}\,(4+\sqrt{-2}) + 1 + \sqrt{-2} \\ 4+\sqrt{-2} &= (2-\sqrt{-2})(1+\sqrt{-2}). \end{aligned}$$

Backward substitution gives

$$\begin{aligned} 1+\sqrt{-2} &= -1+5\sqrt{-2} - \sqrt{-2}\,(4+\sqrt{-2}) \\ &= -1+5\sqrt{-2} - \sqrt{-2}\left[5-4\sqrt{-2}-(-1)(-1+5\sqrt{-2})\right] \\ &= -\sqrt{-2}\,(5-4\sqrt{-2}) + (1-\sqrt{-2})(-1+5\sqrt{-2}). \end{aligned}$$

Problem 2 **8 points**

Find a prime factorization of

$$\alpha = 126 - 84\sqrt{2} \in \mathbb{Z}[\sqrt{2}]$$

and justify your answer.

Solution. We factor

$$\alpha = 126 - 84\sqrt{2} = (3 - 2\sqrt{2}) \cdot 42 = (3 - 2\sqrt{2}) \cdot 2 \cdot 3 \cdot 7.$$

Note that $3 - 2\sqrt{2}$ is a unit, as $N(3 - 2\sqrt{2}) = 3^2 - 2 \cdot 2^2 = 1$. The prime number $3 \equiv 3 \pmod 8$ remains prime in $\mathbb{Z}[\sqrt{2}]$, whereas 2 and 7 can be further factored:

$$2 = (\sqrt{2})^2, \qquad 7 = (3 + \sqrt{2})(3 - \sqrt{2}),$$

where $\sqrt{2}$ and $3 \pm \sqrt{2}$ are primes in $\mathbb{Z}[\sqrt{2}]$. Hence, a prime factorization of α is

$$\alpha = (3 - 2\sqrt{2})(\sqrt{2})^2 \cdot 3(3 + \sqrt{2})(3 - \sqrt{2}).$$

Problem 3 **10 points**

Consider the quadratic ring $R = \mathbb{Z}[\sqrt{17}]$.

(a) [5 points] Prove that $2 \in R$ is a prime.
Hint: Try modulo 4 at some point.

Solution. Let $\alpha = x + y\sqrt{17} \mid 2$ be a nonunit and nonassociate of 2. Then $N(\alpha) = x^2 - 17y^2 \mid N(2) = 4$ implies $N(\alpha) = x^2 - 17y^2 = \pm 2$. Modulo 4, we get $x^2 - y^2 \equiv \pm 2 \pmod 4$. Since $x^2, y^2 \equiv 0$ or $1 \pmod 4$, $x^2 - y^2 \equiv 0$ or $\pm 1 \pmod 4$, meaning that $x^2 - y^2 \equiv \pm 2 \pmod 4$ can never happen. So, $2 \in R$ is a prime.

(b) [5 points] Find two essentially different prime factorizations of $8 \in R$, with different numbers of prime factors.

Solution. Part (a) shows that $2 \in R$ is a prime, so $8 = 2^3$ is a prime factorization of $8 \in R$.

To find another factorization, suppose $\alpha = x + y\sqrt{17} \in R$ has norm $N(\alpha) = x^2 - 17y^2 = 8$. It is easily seen that $(x, y) = (5, \pm 1)$ works. To show that $\alpha = 5 \pm \sqrt{17} \in R$ is prime, suppose that $\alpha = \beta\gamma$, where $\beta, \gamma \in R$ are nonunits and nonassociates of α. Then $8 = N(\alpha) = N(\beta)N(\gamma)$ implies $N(\beta) = \pm 2$ or $N(\gamma) = \pm 2$. But the argument in part (a) shows that neither is possible. Hence, $\alpha = 5 \pm \sqrt{17} \in R$ is prime, implying that $8 = (5 + \sqrt{17})(5 - \sqrt{17})$ is a different prime factorization of $8 \in R$.

Problem 4 **12 points**

(a) *[6 points]* Compute the simple continued fraction expansion of $\sqrt{7}$.

Solution. We start by finding the continued fraction expansion of $\sqrt{14}$:

$$\begin{aligned}
\sqrt{7} &= 2 + \sqrt{7} - 2 \\
\frac{1}{\sqrt{7}-2} &= \frac{\sqrt{7}+2}{(\sqrt{7}-2)(\sqrt{7}+2)} = \frac{\sqrt{7}+2}{3} = 1 + \frac{\sqrt{7}-1}{3} \\
\frac{3}{\sqrt{7}-1} &= \frac{3(\sqrt{7}+1)}{(\sqrt{7}-1)(\sqrt{7}+1)} = \frac{\sqrt{7}+1}{2} = 1 + \frac{\sqrt{7}-1}{2} \\
\frac{2}{\sqrt{7}-1} &= \frac{2(\sqrt{7}+1)}{(\sqrt{7}-1)(\sqrt{7}+1)} = \frac{\sqrt{7}+1}{3} = 1 + \frac{\sqrt{7}-2}{3} \\
\frac{3}{\sqrt{7}-2} &= \frac{3(\sqrt{7}+2)}{(\sqrt{7}-2)(\sqrt{7}+2)} = \sqrt{7}+2 = 4 + \sqrt{7} - 2.
\end{aligned}$$

So $\sqrt{7} = [2; \overline{1, 1, 1, 4}]$.

(b) *[6 points]* Use your answer in part (a) to determine all units of $\mathbb{Z}[\sqrt{7}]$.

Solution. Based on $\sqrt{7} = [2; \overline{1, 1, 1, 4}]$, we compute

			2	1	1	1	4
x	0	1	2	3	5	8	
y	1	0	1	1	2	3	
$x^2 - 7y^2$			-3	2	-3	1	

Since no values of x, y for which $x^2 - 7y^2 = -1$ are found in the table above before $(x, y) = (8, 3)$, which satisfies $x^2 - 7y^2 = 1$, the fundamental unit in $\mathbb{Z}[\sqrt{7}]$ is $8 + 3\sqrt{7}$, and $\pm(8 + 3\sqrt{7})^n$ $(n \in \mathbb{Z})$ give all the units in $\mathbb{Z}[\sqrt{7}]$.

Problem 5 **10 points**

(a) *[5 points]* Suppose that $p \neq 5$ is an odd prime dividing $a^2 - 5$ for some $a \in \mathbb{Z}$. What can you say about p modulo 5? Justify your answer.

Solution. Since $p \mid a^2 - 5$, we have $a^2 \equiv 5 \pmod{p}$. So 5 R p. Given that $5 \equiv 1 \pmod{4}$, quadratic reciprocity gives

$$\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right) = \begin{cases} 1, & \text{if } p \equiv 1 \text{ or } 4 \pmod{5}, \\ -1, & \text{if } p \equiv 2 \text{ or } 3 \pmod{5}. \end{cases}$$

We conclude that $p \equiv 1$ or $4 \pmod{5}$.

(b) *[5 points]* Use numbers of the form $a^2 - 5$ to give a proof that there are infinitely many primes $p \equiv 4 \pmod{5}$. (e.g., 19 is the smallest such prime.)

Solution. Assume to the contrary that there are only finitely many primes congruent to $4 \pmod{5}$, say $p_1 = 19, p_2, \ldots, p_k$. Consider

$$n = (2p_1 \cdots p_k)^2 - 5 = 4(p_1 \cdots p_k)^2 - 5 > 1.$$

Then $2 \nmid n$, $n \equiv 4 \pmod{5}$, and $p_i \nmid n$ for every $1 \leq i \leq k$. By part (a), every prime $p \mid n$ satisfies $p \equiv 1$ or $4 \pmod{5}$. Since a product of numbers congruent to 1 modulo 5 is also 1 modulo 5, n must have a prime factor $p \neq p_1, \ldots p_k$ with $p \equiv 4 \pmod{5}$, contradicting our assumption that $p_1, \ldots, p_k$ are the only primes congruent to $4 \pmod{5}$. Therefore, there are infinitely many primes $p \equiv 4 \pmod{5}$.

Problem 6 (optional bonus) **6 points**

Find all integers x for which $x^4 + 4$ is a prime number p.
Hint: Work in $\mathbb{Z}[i]$.

Solution. Suppose that $x^4 + 4 = p$. Then $p > 4$, so p is odd. Thus, x is also odd. Modulo 4, we see that $p \equiv 1 \pmod 4$.

Working in $\mathbb{Z}[i]$, we have

$$p = x^4 + 4 = (x^2 + 2i)(x^2 - 2i) \in \mathbb{Z}[i].$$

So $x^2 + 2i$ and $x^2 - 2i$ are both prime in $\mathbb{Z}[i]$. Since

$$2i = (1 + i)(1 - i)i = (1 + i)^2,$$

$x^2 - 2i$ (and of course also $x^2 + 2i$) can be further factored in $\mathbb{Z}[i]$:

$$x^2 - 2i = (x + 1 + i)(x - 1 - i).$$

Thus one of $x + 1 + i$ and $x - 1 - i$ is a unit in $\mathbb{Z}[i]$. Since neither of them is a real number, we have

$$x + 1 + i = \pm i \qquad \text{or} \qquad x - 1 - i = \pm i.$$

Comparing the real parts, we obtain $x + 1 = 0$ or $x - 1 = 0$. It follows that $x = \pm 1$. Plugging this into $x^4 + 4 = p$ gives $p = 5$. Hence, $x = \pm 1$ are the only integers for which $x^4 + 4$ is a prime number.

End of examination